

TUTORIAL REVIEW

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Non-invasive biochemical sensing with AI-driven analytics: a comprehensive review of technologies, applications, and future directions

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Artificial Intelligence (AI) combined with non-invasive biochemical sensing is transforming healthcare monitoring and diagnostics. This review explores the scope of smart sensing systems, highlighting their core components, enabling technologies, design challenges, and potential solutions. It examines key biofluids and their analytes to understand suitable sensing modalities and the adaptations required for different biological environments. The review also discusses methods for functionalizing biosensors to achieve higher sensitivity, improving data analytics, and ensuring data security and system robustness. Beyond technical aspects, it emphasizes the importance of regulatory standards and examines the current status of developments in wearable biochemical sensing devices and the market forces driving their growth. Looking ahead, it highlights the future of biochemical sensing applications where advanced technologies such as quantum sensing, transformer models, and neuro-symbolic AI could significantly enhance performance. With this comprehensive analysis, the review aims to guide researchers, clinicians, and industry professionals in understanding both the present landscape and the future directions of AI-driven, non-invasive biochemical sensing systems.

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1 Introduction

1.1 Evolution of biochemical sensing

The evolution of biochemical sensing has witnessed a remarkable transformation from invasive, laboratory-bound procedures into sophisticated, non-invasive monitoring systems that can be deployed in real-world environments. Traditional diagnostic methods, though highly accurate, often require blood sampling, tissue biopsies, or other invasive procedures that can be uncomfortable for patients and may not be suitable for continuous monitoring while, non-invasive measurements reduce time as well as cost spent on measurements, reduce risks and facilitate good accessibility.^{1,2} The emergence of non-invasive biochemical sensing technologies has opened up new possibilities for healthcare delivery, enabling continuous health monitoring, early disease detection, and personalized treatment strategies.³

The idea of detecting chemical changes electrically dates back to 1906, when M. Cremer demonstrated the presence of potential difference across fluids with different pH values.⁴ This discovery paved the way for the demonstration of reliable pH measurement by F. Haber and Z. Klemensiewicz in 1909⁴ and invention of the oxygen electrode by Leland C. Clark, Jr in 1956.⁵ In spite of these inventions, the first amperometric measurement of glucose was demonstrated by Leland C. Clark, Jr *et al.* in 1962.⁶ The development of biosensors began with this demonstration, which laid the foundation for the development of

numerous other enzymatic biosensors.⁷ The first commercial enzymatic glucose sensor (by Yellow Spring Instrument Company) came into existence in 1975.⁸ In early days, biosensors were primarily designed for clinical laboratory use and required invasive sample collection. The list of inventions that paved the way to recent day smart biosensors is presented in Fig. 1. The advent of microelectronics and nanotechnology in the late 20th century enabled the miniaturization of sensing elements and the development of portable diagnostic devices.^{3,9} In the year 1970, an ion-sensitive field effect transistor was developed by Bergveld¹⁰ and in 1975, Suzuki *et al.* demonstrated the first immunosensor using a dehydrogenase–collagen membrane coating.¹¹ Furthermore, fiber-optic biosensors were developed in 1982¹² and the first surface plasmon resonance (SPR) based immunosensors were demonstrated in 1983.¹³ Ferrocene-mediated glucose sensing developed in 1984 enables device miniaturization as well as detection efficacy.¹⁴ However, the first commercial SPR biosensor came into existence in 1990 (Pharmacia Biocore) and the first handheld blood biosensor was commercialized by i-STAT in 1992.¹⁵ More recently, the integration of advanced materials science, flexible electronics, and wireless communication technologies has made truly wearable and non-invasive sensing systems a reality.

1.2 The promise of non-invasive technologies

Non-invasive biochemical sensing represents a fundamental shift in the way health monitoring and disease diagnosis are approached. By analyzing biomarkers present in easily

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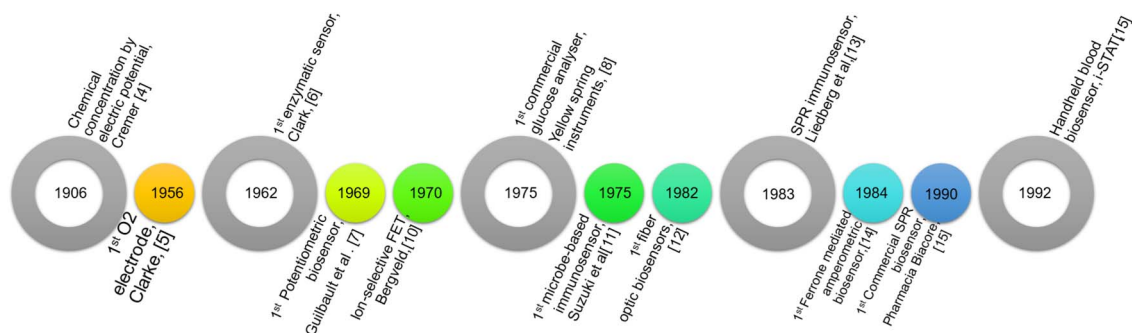


Fig. 1 Evolution of bio-sensors.

accessible biofluids such as sweat, saliva, tears, and exhaled breath, these technologies offer several compelling advantages over traditional invasive methods.^{16,17} First, they eliminate the discomfort and potential complications associated with invasive procedures like blood draws and biopsies. Such a non-invasive procedure gains good patient acceptance and enables frequent monitoring possible. Second, they support continuous or near-continuous monitoring, providing a dynamic view of physiological processes rather than discrete measurements.¹⁸ Third, they can be integrated into wearable devices, allowing for monitoring during normal daily activities and providing insights into how lifestyle factors affect health.

The principle behind non-invasive sensing lies in the fact that most of the biomarkers present in blood also appear in other body fluids, albeit often at different concentrations and with varying time delays.^{19,20} For example, glucose levels in the interstitial fluid (ISF) correlate well with blood glucose, making continuous glucose monitoring through minimally invasive sensors possible. Similarly, stress hormones like cortisol can be detected in saliva, sweat, and tears, providing insights into physiological and psychological stress states without blood sampling.^{21,22}

Though biochemical sensing techniques offer promising results, their predictions are often affected by interference, sensor drifts, and individual physiological differences. Furthermore, using biochemical sensors with wearable (implantable) devices is often affected by (micro) environmental variations. These effects directly impact the biochemical sensors' accuracy and precision. However, the integration of AI techniques could offer beneficial solutions²³ and this will be introduced in the next section.

1.3 AI integration in biochemical sensing

The integration of artificial intelligence and machine learning algorithms has been transformative for biochemical sensing technologies, addressing many of the traditional limitations of sensor-based diagnostics.^{24,25} Raw sensor signals are often noisy, susceptible to interference, and may not directly correspond to clinically relevant biomarker concentrations. AI-driven analytics provide sophisticated signal processing capabilities, pattern recognition, and predictive modeling that can extract meaningful information from complex, multivariate sensor data.^{26,27}

Machine learning algorithms excel at identifying subtle patterns in sensor data that may not be apparent to human observers or traditional analytical methods. They can compensate for sensor drift, environmental variations, and individual physiological differences, improving the accuracy and reliability of biosensor measurements.^{23,28} Furthermore, AI enables the development of personalized models that adapt to individual users over time, potentially improving diagnostic accuracy and reducing false alarms.²⁹ Deep learning approaches, in particular, have shown promise in processing complex, high-dimensional sensor data and identifying biomarker signatures associated with specific disease states.^{30,31}

1.4 Scope and objectives

This comprehensive review aims to provide a thorough analysis of the current state and future prospects of non-invasive biochemical sensing with AI-driven analytics. We examine the technological foundations, clinical applications, and commercial potential of these integrated systems. The review covers key sensing modalities including sweat-based sensors, salivary diagnostics, tear fluid analysis, and breath monitoring technologies. We analyze the role of artificial intelligence in enhancing sensor performance, enabling real-time analytics, and supporting clinical decision-making. There exist numerous reviews on biosensors as well as AI tools for biomedical applications individually. However, studies that discuss the fundamental biochemical sensing principles together with the constituent technologies that drive developments in AI-based biochemical sensing are limited in the literature. Thus, this review addresses this gap by presenting a novel narrative discussing (i) the fundamentals of biochemical sensors, (ii) wearable and continuous health monitoring devices that are explored for sensing analytes from sweat, tear, breath, saliva and their combinations, (iii) the primary technologies that drive the development of biochemical sensing systems and (iv) the future perspectives.

Specific objectives of the review include (1) reviewing the fundamental principles and technological components of non-invasive biochemical sensing systems; (2) examining the integration of AI and machine learning algorithms in biosensor data analysis; (3) evaluating current and emerging applications in healthcare and wellness monitoring; (4) assessing the clinical

validation and regulatory pathways for these technologies; (5) identifying key challenges and limitations that must be addressed for widespread adoption; and (6) exploring future directions and emerging technologies that will shape the next generation of AI-driven biochemical sensing systems.

2 Fundamentals of non-invasive biochemical sensing

2.1 Principles of non-invasive detection

Non-invasive biochemical sensing relies on the detection and quantification of target analytes present in accessible biological fluids without requiring invasive procedures. The fundamental principle underlying these technologies relies on three facts: (i) biomarkers present in blood are also found in other biofluids,^{32,33} (ii) the concentrations and temporal dynamics of biomarkers differ between these fluids and blood, (iii) there exist a correlation between the presence/concentration of these biomarkers and particular disease existence/severity respectively. Thus, decoding these correlations that naturally exist enable clinically meaningful measurements.^{34,35}

Biosensor based non-invasive healthcare monitoring systems consist of five major blocks namely (i) signal sensing and transduction device – the biosensor, (ii) pre-processing and signal conditioning block, (iii) data analytics block, (iv) communication modules and (v) display block. The data analytics block can further be divided into two: the computing platform and method used for computing. The basic block diagram of an AI-driven biosensor system along with the various methods adapted in each block is given in Fig. 2. Each of these blocks is discussed in the subsequent sections below.

Each of these blocks can be realized by employing different principles. The orange boxes in Fig. 2 represent the various techniques that can be adapted for the implementation of each block. For instance, signal sensing and transduction can be carried out using the electro-chemical, optical or mechanical

sensing principle. Pre-processing of received signals is normally carried out on-chip, that is, on the device to prevent transmission losses, while offline smart pre-processing can also be carried out in the case of calorimetric readouts. Similarly, computational algorithms can be developed following conventional rule-based methods, machine learning or deep learning methods and these algorithms can run on either embedded chips on-device, on-edge, on-fog or on-cloud. Furthermore, communication with the computing platform can be enabled by the Bluetooth low energy or Wi-Fi and the measured quantities can be reliably found on display on-device or with the help of smart devices. Each of these blocks in turn is realized from its constituent blocks for enabling successful biochemical sensing, that is, the signal sensing and transduction block consists of a biofluid collection unit, a sample pre-processing stage, a platform for analyte recognition and binding – the actual bio-receptor, and signal transduction. The pre-processing block has filters and multiple stages of amplifiers. Each of these blocks must be optimized for reliable and accurate measurements with non-invasive measurements.³⁶ The choice of these methods depends on the biofluid used for sensing, the target analyte being sensed, the required sensitivity, required time of response, the type of analysis, device complexity and cost of the device.^{37–39} Furthermore, the computational methods used for processing and analytics depend on the type of prediction. These methods use rule-based computations or feature extraction-based predictions depending on the requirements.

2.2 Biofluids as non-invasive sensing platforms

Biofluids being considered for non-invasive sensing carry importance as they are the medium which carries the analyte of interest to be sensed. Each of the biofluids carries traces of various analytes that are present in blood, however, certain analytes are rich in particular biofluids and thus, knowing them and selecting the particular sensing medium carries significance. For instance, sweat, which is an easily accessible

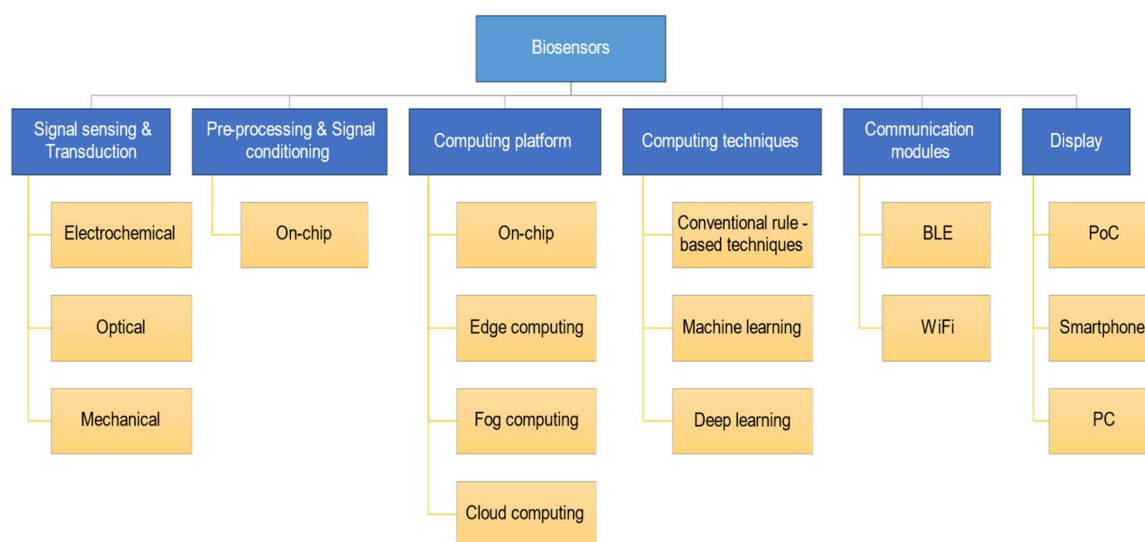


Fig. 2 Blocks in AI-driven biosensor systems.

Table 1 Constituents in sweat, saliva and tear

Sensing medium – biofluid	Ions	Metabolites	Drugs	Biomolecules	Ref. No.
Sweat	Na ⁺ , Cl [−] , K ⁺ , Ca ²⁺ , NH ₄ ⁺ , Zn ²⁺ , Cd ²⁺ , Pb ²⁺ , Cu ²⁺ , Hg ⁺ , pH	Glucose, lactate, uric acid, ascorbic acid, cortisol	Levodopa, caffeine, alcohol	Peptides	40
Saliva	K ⁺ , Na ⁺ , Ca ²⁺ , Mg ²⁺ , Cl [−] , F [−] , HCO ₃ [−] , H ₂ PO ₄ [−] , HPO ₄ ^{2−}	Alpha amylase, glucose, lactate, uric acid, amino acids, lipids	—	C-reactive protein (CRP), cardiac troponin T (hs-cTnT), alpha-2-HS glycoprotein, NT-proBNP, alpha-2 macroglobulin, IL-1beta, IL-6, TNF-alpha prostaglandin E2, CYFRA-21-1, IgA, Nisin	45 and 134
Tear	Na ⁺ , K ⁺ , Cl [−] , and bicarbonate	Glucose, urea, cortisol	—	Lysozyme and lactoferrin, immunoglobulin A (IgA) and IgG, albumin and globulins	47 and 48

biofluid, carries numerous clinically relevant analytes and enables promising non-invasive sensing solutions.^{40,41} Electrolytes like sodium, potassium, and chloride, metabolites like glucose, lactate, and urea, and various biomolecules representing the systemic physiological states are present in sweat. A list of various analytes present in sweat is given in Table 1. Wearable sweat sensors have been successfully used for monitoring the hydration status, metabolic state, electrolyte balance as well as the level of drugs.^{42,43} However, sweat collection and analysis present unique challenges, including low analyte concentrations, potential contamination from the skin surface, and the need to stimulate sweat production in some applications.⁴⁴

Considering saliva, it contains numerous analytes ranging from proteins, nucleic acid, hormones and small molecules present in blood and thus offers an attractive platform for non-invasive biochemical sensing.^{45,46} A list of various analytes present in saliva is given in Table 1. Due to the presence of numerous biomarkers, saliva enables reliable detection of stress, inflammations, cancer, and even genetic disorders. The primary advantage in using saliva-based sensing relies in its easy accessibility, availability and presence of higher percentages of analytes compared to sweat and tear. However, a primary limitation with the use of salivary biomarkers is that the oral biochemical sensing systems are prone to frequent biofouling and bacterial infections. Next in the pipeline, tear is analyzed as a promising biofluid for various diagnoses and ocular health monitoring. A list of various analytes present in tear is given in Table 1. Compared to other non-invasive fluids, tear is available in smaller quantities but contains numerous

metabolites, proteins and other analytes and thus offers a reliable platform for biochemical sensing. In particular, tear-based glucose monitoring and diabetic diagnosis have been very successful.^{47,48} However, bio-comfort poses a major problem to tear sensors.

Exhaled breath contains numerous volatile organic compounds (VOCs), other gases and small volumes of non-volatile condensates and thus represents a distinctive non-invasive sensing modality.^{49,50} More than 1000 VOCs originating from various metabolic processes were found in exhaled breath facilitating a promising platform for detecting metabolic disorders, respiratory diseases and even certain types of cancer.⁵¹ The primary advantage of adapting breath analysis relies in their true non-contact sampling and the ability to use pulmonary excretion for reliable health monitoring. A list of various analytes present in exhaled breath is given in Table 2.

In spite of its promising potential, a primary limitation with breath analysis resides in the very low concentration of many target analytes, and environmental interferences. This necessitates adequate sample conditioning and sophisticated analytical techniques to achieve good sensitivity and specificity.^{52,53} The challenges that appear with the use of biochemical sensors can be reliably addressed when a biochemical sensing system is integrated with AI-based analysis. These methods establish correlation between the sensed signal and the analyte concentration which can be calibrated and personalized for specific needs. Thus, the integration of AI methods enables overcoming the limitations with biochemical sensing systems. The primary components that make up the biosensor probe or device, and the biofluid-specific modification to be considered while

Table 2 Constituents of breath

Sensing medium	VoCs	Inorganic gases	Non-volatile compounds	Diseases diagnosed	Ref. no.
Breath	Acetone, isoprene, ethane, pentane	CO ₂ , CO and NO	Peroxonitrite, cytokines, and isoprostanes	Lung diseases, gastro intestinal disorders, cancers, Alzheimer's and Parkinson's disease	50

designing for effective biorecognition, are discussed in Section 4. However, the below section discusses the bioreceptors and types of transduction mechanisms frequently used in a biosensor device.

2.3 Sensing & transduction mechanisms

A biosensor as a device consists of two components, namely, (i) a bioreceptor, and (ii) a transducer. Bioreceptor is a biological element that recognizes a target analyte and responds to it by producing a signal while the transducer picks up this biological response and converts it into a measurable signal depending on the transduction principle. A bioreceptor produces charges, chemical change, light, or heat in response to the biorecognition event. Different types of bioreceptors are shown in Fig. 3(a). It could be inferred from Fig. 3(a) that antibodies, enzymes, cells, aptamers, nucleic acids and nanoparticles act as bioreceptors. Specifically, antibodies and aptamers can detect specific proteins and hormones reliably, enzymes can detect metabolites, and nucleic acids can reliably detect DNA, RNA of viruses, bacteria and other microorganisms. The conversion of biochemical recognition events into measurable signals requires sophisticated transduction mechanisms that can operate reliably in biological environments.⁵⁴ It is inferred from Fig. 3(b) that there are three most frequently used transduction mechanisms in the case of biochemical sensors. Furthermore, they are sub-divided based on the output produced by them. Specifically, electrochemical sensors can be amperometric (current output), voltametric (produce voltage as output) or impedimetric (produce difference in resistance). Similarly, the optical output may be in the form of fluorescence, surface plasmon resonance (SPR), emission, absorption or scattering. Mechanical sensors may be based on crystals, acoustic wave or thin cantilevers to convert mechanical forces to an electrical quantity.

Electrochemical transduction is the most frequently used technique due to its high sensitivity, faster response, miniaturization capabilities and bio-compatibility. Electrochemical biosensors convert the biorecognition event (either analyte binding to bioreceptor or enzymatic reaction) into measurable electrical quantity say current, potential or impedance.^{55,56} Based on the converted quantity, biosensors are classified as amperometric, potentiometric, or impedimetric devices.

Optical transducers leverage changes in light absorption, emission, scattering, or refractive index to detect target analytes.^{57,58} Among them, fluorescence-based biosensors are frequently used due to the availability of diverse fluorescent labels and probes and high sensitivity. Similarly, surface plasmon resonance (SPR) sensors are the next frequently used biosensors and they quantify the analyte concentration based on the refractive index of light resulting from biorecognition events. These SPR sensors are also known for their high sensitivity. In addition, they enable label-free detection. Wearable devices that use optical transduction mechanisms use miniaturized light sources, detectors, and optical components for their realization, though miniaturization poses certain difficulties.^{59,60}

Mechanical transduction relies on detecting changes in mass or mechanical properties of the sensing element in order to quantify the biorecognition event.⁶¹ Most frequently used biosensors based on mechanical transduction principles are quartz crystal microbalance (QCM) and surface acoustic wave (SAW) sensors. In addition, cantilever-based sensors that are capable of detecting mechanical deflections resulting from analyte binding over bioreceptors are also used adequately.⁶² All of these sensors offer highly sensitive measurements that can quantify even single molecule binding as well as facilitate faster response for real-time monitoring capabilities. However,

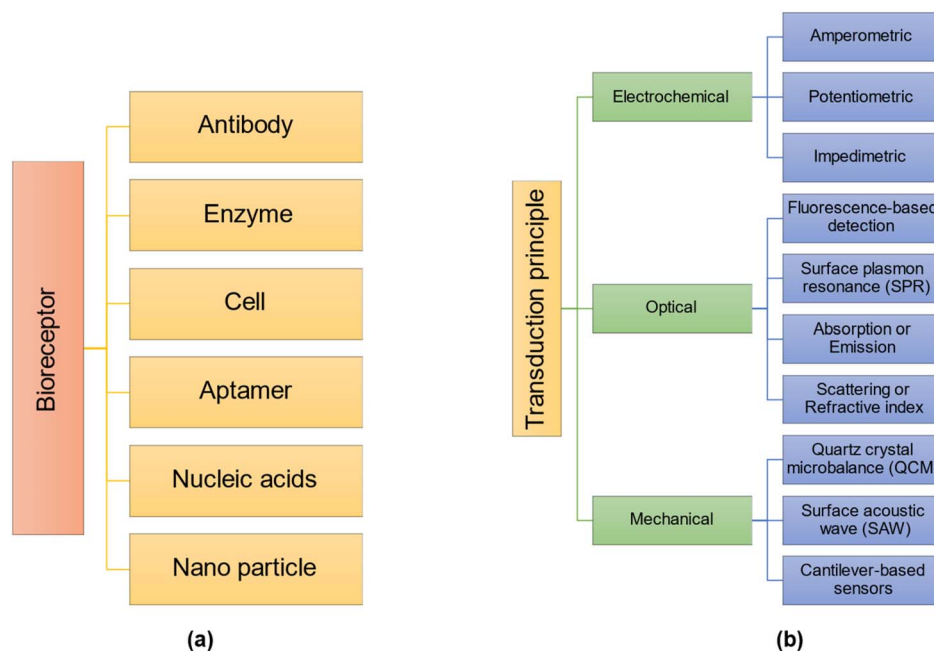


Fig. 3 Biosensor probe and its constituents. (a) Bioreceptors, (b) transduction principles.

a major challenge with the use of mechanical sensors is that they are often affected by noise and interference from the environment and thus, demand careful isolations from mechanical vibrations and temperature fluctuations.^{63,64}

2.4 Signal processing fundamentals

Raw signals as received from the biosensor element are weak as well as often corrupted by noise and interference. This necessitates adequate processing of raw signals in order to extract meaningful biochemical information from them. Furthermore, data analytics from sensed signals demands extraction of features from them.^{65,66} Thus, a signal processing block includes numerous stages which include noise reduction, signal conditioning, feature extraction, and calibration.

Noises are inherent in sensed bio signals and they often originate from motion artifacts, biological variability, thermal effects and electromagnetic interference. Noise reduction is crucial in non-invasive sensing applications where signal levels are relatively low and subject to numerous interferences.⁶⁷ Signal amplification, filtering and analog-to-digital conversion units constitute the signal conditioning stage. And the signal resulting from this stage are ensured to be appropriate for further digital processing. Advanced filtering techniques, like adaptive filters and digital signal processing algorithms, help minimize noisy signals at the same time preserving the underlying biological signal. In addition to noise removal, baseline correction and drift compensation are also carried out for long-term monitoring applications where sensor characteristics may change over time.^{68,69}

Once the measured signals are conditioned, meaningful features should be extracted from them. This facilitates identifying/quantifying specific signal characteristics that correlate with target analyte concentrations or biological states. Examples for such features include (i) peak intensity/area in spectroscopic measurements, (ii) statistical properties of signals, (iii) frequency domain analysis of time-varying signals.^{70,71} Once the necessary correlation is established, the device should be calibrated. Calibration involves establishing a relationship between sensor signals and actual analyte concentrations. This step often requires multiple standards and accounting for matrix effects and individual variations.

Calibration facilitates a meaningful user interface for biosensor users. The above-mentioned block together with a rule-based computing algorithm is sufficient for the realization of a biosensor-based point-of-care diagnosis device. However, continuous analyte monitoring, disease predictions, and anomaly forecasting demand data analytics and use of artificial intelligence-based learning algorithms. Furthermore, machine learning approaches are also used in the signal conditioning stages in order to automate as well as improve the signal processing stage.^{72,73} With this brief introduction to the fundamental blocks in a biosensor based non-invasive sensing system, the four major technologies that drive developments of non-invasive biochemical sensing system namely (i) artificial intelligence algorithms, (ii) biofluid specific sensor design, (iii) advanced materials and nanotechnology and (iv) computing

platforms for AI-based data analytics are discussed in the upcoming sections. An elaborate discussion of various machine learning algorithms for biosensor data analytics is presented in the below section.

3 AI-driven analytics in biochemical sensing systems

3.1 Machine learning (ML) algorithms in biosensor data analysis

Various AI techniques that are reliably used for non-invasive sensing analysis are given in Fig. 4. AI techniques are broadly classified into two: machine learning (ML) and deep learning (DL). Machine learning techniques require a separate feature extractor block for extracting features from the input and use a ML classifier for making predictions while DL methods can extract features by themselves. Machine learning algorithms are capable of extracting meaningful patterns from complex, multi-dimensional sensor data and thus provide sophisticated biosensor data analytics.^{74,75} Conventional techniques rely on rule-based arithmetic, simple mathematical models and pre-determined calibration curves for capturing the full complexity of biological systems. Machine learning algorithms use either supervised or unsupervised learning as depicted by the first two blocks of Fig. 4. Deep learning algorithms are categorized as CNN – that work well with image inputs, RNN – excellent in working with time series inputs and transformer models that are efficient in attention based learning. However, both these techniques can adapt to non-linear relationships and are capable of identifying subtle patterns from data that may not be apparent through conventional analysis.

Supervised learning based ML algorithms use labelled training data to understand their inherent patterns and thus readily used in biosensor applications where labelled training data are available. Among them, linear and polynomial regression models use either linear or polynomial equations to fit the relationship between dependent and independent variables. These are the simplest and valuable methods in establishing the baseline relationship between sensor signals and analyte concentrations.⁷⁸ Support vector machines (SVMs) represent data in 'n' dimensional space and find the optimal hyperplane that best separates the various classes of data. SVMs are effective in classification tasks such as distinguishing healthy and diseased states from biomarker profiles.^{76,77} Random forests and gradient boosting algorithms are ensemble-based learning methods that offer robust performance for both classification and regression tasks. In addition, they can handle noisy data well and provide feature importance rankings that can guide sensor design and optimization.⁷⁹

Unsupervised learning techniques are capable of finding meaningful patterns and relationships within unlabelled data. Such learning methods play a significant role in exploratory analysis and pattern discovery from biosensor data.⁸⁰ Clustering algorithms such as k-means clustering and hierarchical clustering reveals the natural groupings that exist in data. These groups may correspond to different physiological states or

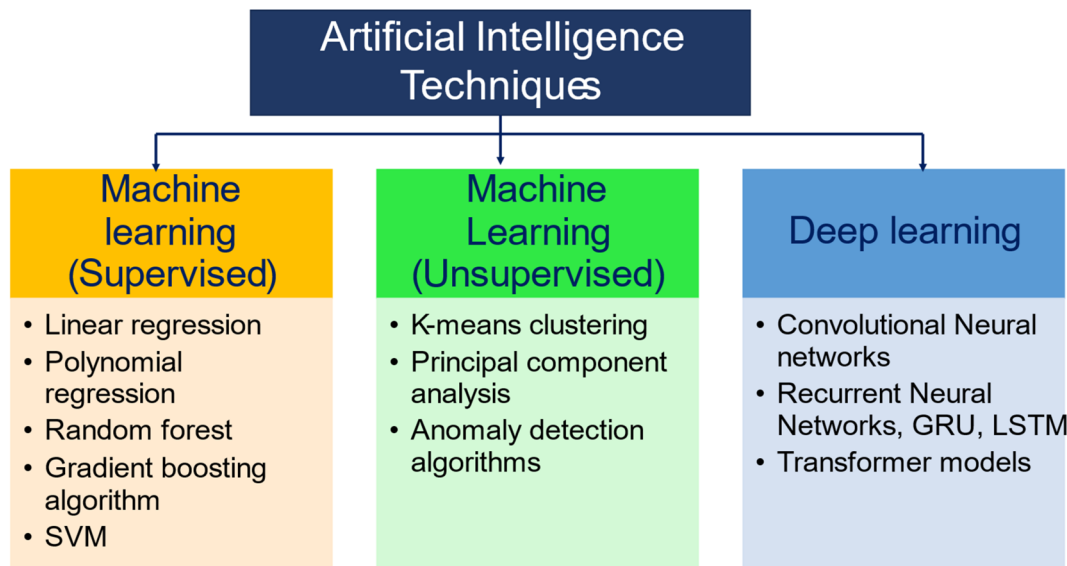


Fig. 4 AI techniques used for biosensor analytics.

patient populations. Similarly, principal component analysis (PCA) is another method that adapts dimensionality reduction methods to visualize high-dimensional sensor data and identify the most informative features.⁸¹ Furthermore, anomaly detection algorithms aid in identifying unusual sensor readings and thus enable identifying equipment malfunction, sample contamination or clinically significant events requiring immediate attention.^{82–84} These unsupervised algorithms are helpful in cases where there is limited labelled data or labelled data are expensive to obtain. Thus, these algorithms are often used with clinical applications.

3.2 Deep learning (DL) for pattern recognition

Deep learning algorithms have emerged as a powerful tool for retrieving patterns from complex biosensor data, specifically, time-series measurements, multi-modal sensor array data and high dimensional spectroscopic data.^{85,86} Convolutional neural networks (CNNs) are capable of learning hidden features in biosensor data and extract them automatically without manual feature engineering. Due to their remarkable capabilities, CNNs can process spatial or spectral structured data, from optical sensor arrays or spectroscopic measurements reliably. Thus, CNNs are particularly attractive for biosensor applications where optimal features may not be obvious.^{87,88} Another class of DL algorithms are based on recurrent neural networks (RNNs), having the ability to model temporal dynamics which are efficient in recognizing physiological processes that evolve over different time scales, from seconds to days to weeks. Such a capability enables capturing long-term dependencies in time-series data, which is very helpful in continuous monitoring applications. In the context of healthcare, such examples include continuous glucose monitoring and cardiac function monitoring or tracking the progression of biomarker levels over time. Two types of RNN architecture frequently used for

analyzing temporal patterns in biosensor data are Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU).^{89,90}

Transformer architecture, originally developed for natural language processing, has recently been adapted for biosensor data analysis with promising results.^{91,92} The attention mechanism in transformers can identify relevant patterns across different time scales and sensor modalities, potentially improving the analysis of multi-sensor systems. Hybrid architecture that combines different deep learning approaches is also being explored, such as CNN-LSTM networks that can capture both spatial and temporal features in sensor data.⁹³ The success of deep learning in biosensor applications has been facilitated by the availability of large datasets, improved computational resources, and the development of specialized architecture designed for biomedical data.^{94,95}

3.3 AI-based signal processing and noise reduction

AI-based algorithms are frequently used for data analytics and decision support in healthcare industry, however they excel in noise reductions and signal processing tasks too. Specifically, AI-driven signal processing techniques could extract clean and meaningful signals from noisy biosensor data.^{96,97} Conventional signal processing techniques use filters of fixed design (pre-determined parameters) that are not able to adapt to varying noises and individual differences. However, machine learning-based signal processing can adapt to changing conditions, due to their learning capabilities. Adaptive filtering algorithms based on machine learning principles can learn features from the noisy input and adjust filter parameters based on signal characteristics.⁹⁸ Such an AI-based approach enables effective noise removal while preserving significant biological information. For instance, adaptive filters are efficient in removing motion artifacts from wearable sensors in spite of maintaining the physiological signals of interest. Kalman filters and particle filters when enhanced by machine learning algorithms could

provide optimal state estimation even in the presence of noise and uncertainty.^{99,100} Similar to ML techniques, even DL models could support signal processing and complex denoising tasks.^{101,102} For example, autoencoders can learn clean and noisy representations and from learned knowledge could reconstruct clean signals from noisy inputs. Variational autoencoders are a variant of encoder models that use probabilistic approaches for reconstructing signals from noisy signals.¹⁰³ These encoder-based models use generative adversarial networks (GANs) to generate synthetic clean data for training denoising models when such clean data are not available. These adaptive filtering mechanisms could achieve significant results when traditional methods fail.^{104,105}

3.4 Predictive analytics and decision support

Predictive analysis of biosensor data could enable (i) proactive healthcare, (ii) personalized medicine, (iii) risk stratification and (iv) decision support.^{106,107} Proactive and personalized health care is associated with predicting disease states earlier utilizing present medical reports and medical history in a patient-specific manner. AI-based methods, in addition to their reliable prediction of current medical status based on biomarker data, are also capable of analyzing trends and predict future health status.^{108,109} This enables a reliable decision support to healthcare providers and patients. In particular, time series forecasting models can predict future biomarker trajectories based on current and historical data which enable early medical interventions even before clinical symptoms appear.^{110,111}

Risk stratification algorithms play a significant role in assessing likelihood of adverse effects and disease progression. These algorithms are efficient in learning patterns from multiple sensors and external factors like lifestyle, genetics, and environmental conditions, and are capable of providing comprehensive risk assessments.^{112,113} Furthermore, ensemble based learning methods could make predictions based on results from multiple models thus providing enhanced accuracy and reliable predictions.^{114,115} As false predictions, both false positives and false negatives could create undesirable consequences in the healthcare scenario, and quantifying uncertainties in AI predictions becomes crucial for clinical applications.

Decision support systems (DSSs) are built on top of biosensor data and are capable of providing actionable recommendations for healthcare management.^{116,117} These systems can suggest lifestyle modifications, medication adjustments, or the need for additional testing by learning sensor data patterns. DSSs that use reinforcement learnings aid in optimizing treatment plans and thus enable patient outcome over time.¹¹⁸ In addition, DSSs for healthcare applications are often recommended to be integrated with explainable AI techniques in order to provide reasoning for AI recommendations.^{119,120} Furthermore, Natural Language Processing (NLP) integrations enable presentation of complex analytical results in understandable formats for both healthcare providers and patients.

Having been introduced to the fundamental signal sensing, processing and AI driven analytics of biosensor data, the below

discussions highlight biosensor system-specific modifications based on the sensing platform (biofluid), biofunctionalization by nanotechnologies for specific analytes, type of computing and security necessities for the system. The fundamental principle of biochemical sensing and the basic types of AI techniques were discussed in the sections above. The below sections discuss the recent research work related to wearable and continuous health monitoring devices.

4 Wearable and continuous monitoring technologies

Non-invasive sensing from biofluids (such as sweat, saliva, tear and breath) enables disease diagnosis and healthcare monitoring as well as aids in personalized treatments. Successful identification of biomarkers and diagnosis of various anomalies from non-invasive sensing principles and the challenges that exist with biofluid sensing are presented in Fig. 5.

The top row in Fig. 5 represents the different biofluids from which non-invasive biochemical sensing is made. Each element in the first row is divided into two columns with the left column presenting the challenges that limit reliable sensing and the right column presents the biomarkers detected and diseases diagnosed. The below section presents the recent developments in AI enabled biochemical sensing systems under four topics, namely, sweat, saliva and tear, breath and multi-modal sensing.

4.1 Sweat-based sensing systems

Sweat offers a promising platform for non-invasive sensing due to its rich composition and easy accessibility through wearable devices. Modern sweat sensor based wearable devices are self-powered and they use various energy harvesting techniques like, tribo-electric nano generators, piezo-electric nanogenerators, thermoelectric nanogenerators, biofuel cells and photovoltaic cells.^{121,122} Furthermore, they enable wireless communication with smart devices or cloud based platform for real-time monitoring and analysis of health data. These enhancements add to the capabilities of the biosensor based non-invasive healthcare monitoring system.

However, there exist numerous challenges inherent to biochemical sensing capabilities which include low analyte concentrations, sample evaporation, and contamination from the skin surface.¹²⁵ Addressing these challenges with the help of appropriate device architecture and advanced materials paves the way for effective biosensor operation. Development of soft microfluidic systems are crucial for facilitating effective sweat collection and analysis directly on the skin surface.^{123,124} Microfluidic patches often include an integrated sweat collection unit (with or without an induced sweat secretion unit), biofluid routing channels (for multiplexed analysis) and the actual biosensor(s), in a single micro-chip. The actual biosensor(s) unit may use different sensing modalities depending on the analyte being sensed. For example, ion-selective electrodes are used for sensing electrolytes, enzymatic sensors are used for sensing metabolites and impedance-based sensors are used for monitoring sweat rate and volume.^{126,127}

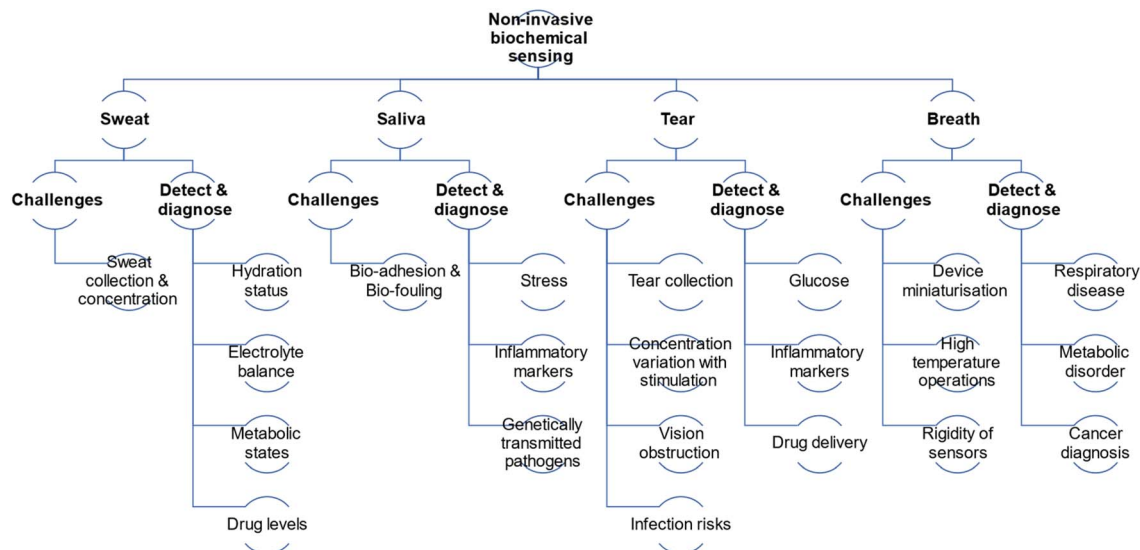


Fig. 5 Biomarker identification and disease diagnosis from biofluids.

Next to the biosensor efficacy lies the computational capabilities and analytics with the acquired data. AI algorithms offer numerous interpretations, correlations and predictions from the sensed data. These algorithms are efficient due to their capabilities of establishing correlations between sweat and blood biomarker levels which are complex and individually variable.^{128,129} Furthermore, ML algorithms can incorporate dependencies from numerous factors such as sweat rate, skin temperature, ambient conditions, and individual physiological differences while making predictions and thus offer more accurate results. In addition, AI algorithms can be used for making predictions on either population-based calibrations or patient-specific manner, thus offering customized or personalized models.¹³⁰ However, this necessitates a synergistic effect from advanced signal processing techniques that aid in compensating sensor drift, temperature effects, and other sources of measurement variability.^{131,132}

4.2 Saliva and tear biomarker detection

Salivary diagnostics could offer reliable oral health monitoring as well as enormous information on metabolites, disease biomarkers and effects of food on oral health. Oral and dental patches were demonstrated to offer reliable predictions of dental carries, orthodontic treatments, dental implant health, oral pH, dental microbiota as well as systemic health conditions.^{133,134} For instance, correlation between salivary concentrations of CRP and IL-6 and their concentrations in serum is established experimentally which enables diagnosis of rheumatic diseases.¹³⁹ Similarly, biomarkers like C-reactive protein and interleukins are identified and demonstrated as reliable biomarkers for diagnosing neonatal sepsis.^{137,138} Oral sensors based on various devices and sensing principles were developed for the prediction of specific analytes from saliva. For instance, highly sensitive graphene field effect transistors (G-FETs) with RNA probes were used for identifying human carbonic

anhydrase 1 (CA1) which is useful in diagnosing cancers, diabetes, pancreatitis and Sjogren's syndrome.¹⁴¹ Similarly, biosensors that use paper based substrates for multiplexed sensing of saliva biomarkers have been developed.¹⁴⁰ Furthermore, 3D printed fluidic devices were developed and demonstrated to offer reliable prediction of inflammatory lung disease.¹⁴² However, challenges exist in the realization of salivary sensors. The primary limitations include (i) bio-adhesion phenomena and (ii) bio-fouling resulting from the microbes in the oral cavity.^{135,136} Bio-adhesion is a phenomena which aids in keeping the oral cavity lubricated with enzymes and biomolecules in order to protect the hard tissues. This results in the formation of biofilm and pellicle deposits over the sensor surface which prevents the analyte from binding to the bioreceptor in the sensor surface as well as causes biofouling. Such a difficulty is rectified by using anatomically correct sensors for use with dental cavity. Sensors in the form of flexible polymer foils, attached to mouth guards/pacifiers, and silk-based dental tattoos were proposed for reliable operations. Integration of AI techniques enables reliable and robust analytics from sensed data.

Tear offers a unique platform for ocular health monitoring and biomarker identification. Tear-based sensing systems enable reliable predictions on dry-eye disease, ocular infections, conjunctivitis, and diabetes mellitus.^{143,144} Major limitations with tear sensing mechanism are (i) tear collection and (ii) variation in the analyte concentration that results from external stimulation. These limitations will be rectified when a lens type tear sensor is used. However, contact lens sensors are often limited by (i) infection risks and (ii) vision obstruction.^{145,146} Interestingly, AI algorithms offer numerous opportunities to compensate for the effects of tear production rate, blinking, and other physiological factors and thus can be reliably used for making predictions with tear-based biosensor data.¹⁴⁸ For instance, skin adhesive, colorimetric tear sensing patch

sensors, along with AI assisted analytics were demonstrated to exhibit reliable biomarker identification from tear fluid. Biomarkers including Ca^{2+} , pH, vitamin C, and proteins are identified by such a tear sensing patch.¹⁴⁷

4.3 Breath analysis technologies

Breath analysis offers a truly non-invasive approach for biochemical sensing by detecting VOCs, other gases and condensates from exhaled breath. There are proofs that depict the use of breath odor for disease diagnosis (by the ancient Greeks).^{149,150} Even recent studies have proven that breath analysis could be reliably used for detecting respiratory diseases, metabolic disorders, liver disease, kidney dysfunction, and certain types of cancer.^{151,152} For instance, VOCs in breath were utilized for diagnosing an emerging environmental disease, the multiple chemical sensitivity (MCS) which results from chronic hypoxia.¹⁵⁰ Modern breath sensing based systems employ various detection technologies like gas chromatography-mass spectrometry (GC-MS), electronic nose sensors, and optical spectroscopy for breath analysis.^{153,154} Electronic nose systems use an array of sensors with a varied range of selectivity patterns that facilitate creating a unique fingerprint for different breath compositions. These systems are integrated with AI analysis and pattern recognition algorithms in order to facilitate automatic recognition of even subtle differences in sensor array responses that correlate with different health conditions.^{155,156} Specifically, breath sensors and breath analysis by AI have been transformative for identification of complex biomarkers that enable the diagnosis of chronic kidney disease (CKD), chronic obstructive pulmonary disease (COPD), and lung cancer.¹⁵⁷ For instance, ML methods like linear regression and SVMs were demonstrated to provide predictive analysis based on breath data¹⁵⁸ while DL methods like long short term memory and attention mechanisms were demonstrated to offer reliable prediction of respiration rate based on biosensor data.¹⁵⁹ In particular, DL approaches enable promising predictions. For instance, one dimensional CNN, when used for predicting lung cancer, was demonstrated to achieve an accuracy of 97.8%.¹⁶⁰ Similarly, indium-oxide-based sensors are used for identification of VOCs in artificial breath, and ML based analysis for diagnosis of raised lipid metabolism rate was demonstrated to achieve robust outcomes from non-invasive breath analysis.¹⁶¹ Recent developments in AI-enabled breath analysis devices have the potential to offer a vast range of diagnosis and monitoring capabilities for both point-of-care as well as remote monitoring setups.^{162,163} In spite of their capabilities, major limitations with breath sensing systems rely on numerous factors like (i) the rigid nature of breath sensors, (ii) their reliance on high temperatures for proper operation, and (iii) difficulties with device miniaturization.⁵⁰

4.4 Multi-modal sensing platforms

Next generation non-invasive biochemical sensing will facilitate an integrated sensing platform that offers sensing multiple as well as multimodal signals which facilitates a multiplexed diagnosis. Such a system could be realized from (i) monitoring

numerous biomarkers with a single device, (ii) measuring the same biomarker with different sensing mechanisms in order to provide enhanced reliability and comprehensive health assessments.^{164,165} For instance, electrochemical and optical sensing platforms which are adequately investigated for understanding metabolic syndrome (MS) can be combined to sense multiple analytes for enabling reliable prediction of MS states.¹⁶⁶ Similarly, bio-phonic sensors, electrochemical sensors, motion and temperature sensors are used for sensing various biomarkers like sweat lactate, pH, tissue oxygenation, heart rate, motion, and temperature in order to understand various physiological markers during a high intensity exercise protocol.¹⁶⁷

Frequently used modalities in healthcare applications include medical images, biosensor data, genomic data, environmental data, behavioral data, and electronic health records. Fusing these modalities offers reliable and robust healthcare decisions and predictions.¹⁶⁸ Development of multi-modal sensing platforms offers unique challenges in terms of system integration, data fusion and user-interface design. These challenges exist with rule-based conventional computing algorithms, however, they could be addressed using AI enabled algorithms based on machine learning and deep learning. AI plays a crucial role in multi-modal data fusion, extraction of complementary information from various sensors and thus facilitating more robust and accurate diagnosis.^{169,170} There are different types of learning techniques available which include multi-view learning, tensor-based consensus learning, attention mechanisms *etc.*

Multi-view learning techniques enable learning the features from different views of the same data, that is, inputs take different forms say text, images, and numeric, and features are learnt from all of these inputs. Predictions are made based on all learned features. Such learning techniques provide a deeper insight into the complex features of a system.¹⁷¹ Furthermore, tensor-based learning techniques were widely investigated for overcoming the limitations with multi-view learning techniques. For instance, the tensor-based consensus learning technique was demonstrated to overcome limitations with missing views in incomplete multi-view clustering.¹⁷² In addition, attention mechanisms that automatically weigh the contributions of different sensors based on their reliability and relevance to specific diagnostic tasks are significant for multi-modal sensing. For instance, multi-modal depression attention models were demonstrated to extract features from audio, visual and textual modality and were successfully used for detecting inter- and intra-modality features for effective identification of depression.¹⁷²

Furthermore, the integration of edge computing capabilities enables real-time processing of multi-modal sensor data, reducing latency and improving user experience.^{173,174} With these present enhancements and developments in progress, future biochemical sensing systems will be able to provide increasingly comprehensive and personalized health monitoring capabilities, which potentially replace multiple separate sensing units with a single integrated wearable system.^{175–177}

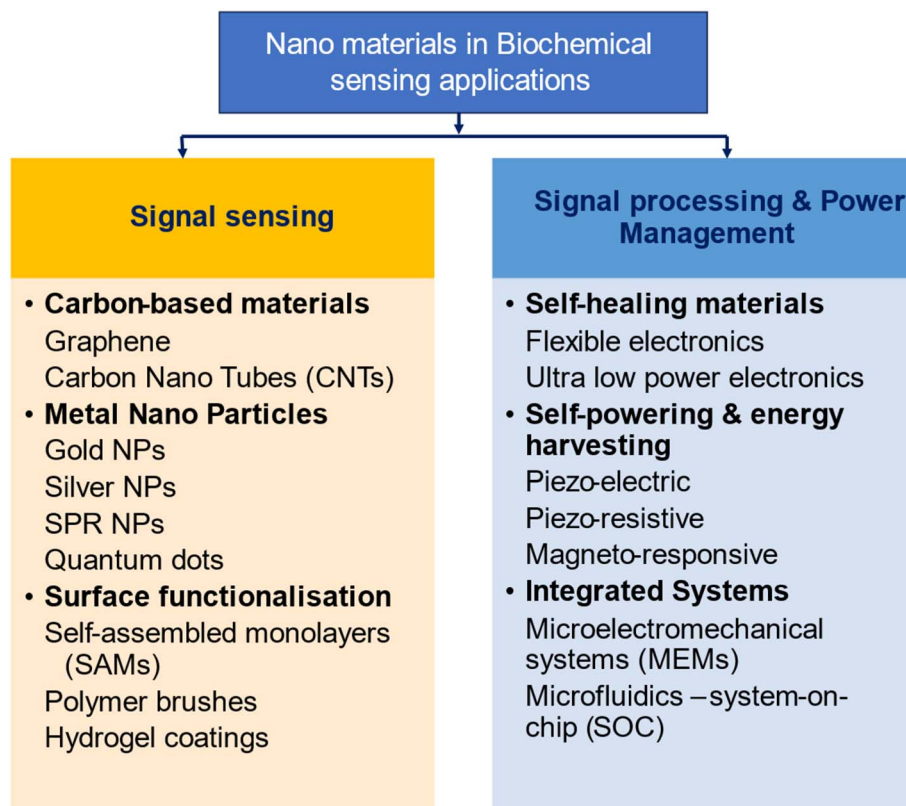


Fig. 6 Nanomaterials used for biochemical sensing.

5 Advanced materials and nanotechnology integration

Effective non-invasive sensing of analytes from biofluids for healthcare applications is not devoid of challenges. The challenges that exist while using different biofluids as sensing medium are presented in Fig. 3 and discussed in the previous section. These challenges are almost always addressed in two ways: (i) AI-based analytics which rectifies limitations by appropriate correlation measurements (however, it should be ensured that AI-based optimizations are devoid of any form of bias, which will be discussed in Section 8), (ii) reliable and accurate sensing, flexible materials, device miniaturization, biocompatibility, bio-comfort and similar material related challenges are potentially addressed by developments in nanotechnology and discovery of nanomaterials. A list of nanomaterials used for biochemical sensing applications is given in Fig. 6. Thus, the integration of nano materials for bio-sensing applications plays a significant role in advancing non-invasive sensing systems and thus is discussed in detail in the below section.

5.1 Nanomaterials in biosensor design

Nanomaterials, defined as materials with at least one dimension in the nanoscale, are known to possess distinct properties due to the nanoscale nature, which include higher surface-to-volume ratios, quantum effects, and tunable surface

chemistry. The integration of nanomaterials with biosensors offers dramatic improvements in sensitivity and selectivity of biomarkers as well as facilitate device miniaturization, flexibility and biocompatibility.^{178,179} Nano biosensors offer such a performance through improved signal transduction, increased binding capacity and enhanced material stability.^{180,181}

Carbon-based nanomaterials are adequately investigated for bio-medical applications due to their excellent electrical conductivity, large surface area, and biocompatibility. Specifically, carbon nanotubes, graphene, and carbon quantum dots find widespread application in biosensor developments.^{182,183} Graphene with its distinct crystalline structure with a single element, offers numerous benefits like cost effectiveness and good sensing accuracy. Furthermore, carbon-based sensors enable detection of analytes at extremely low concentrations. Such a property is demanding for early disease detection applications.¹⁸⁴ Similarly, carbon nanotubes (CNTs) when functionalized with specific recognition elements could function as highly sensitive biosensors for various biomarkers. CNTs provide binding sites for bioreceptors as well as transduce the signals generated by a biorecognition event and thus offer a one-step solution for sensing and transduction.¹⁸⁵ Furthermore, the exceptional mechanical properties of carbon nanomaterials enable development of flexible and stretchable sensors for wearable applications. Next to carbon-based nanomaterials, metal-based nanomaterials are the widely investigated ones.

Specifically, gold and silver nanoparticles were adequately investigated for the development of biochemical sensors. And their distinct optical properties are effectively used for highly sensitive biochemical sensing.¹⁸⁶ For instance, Ag NPs and Au NPs are used adequately for improving Surface Enhanced Raman spectroscopy (SERS) signals resulting from a biorecognition event and thus enable reliable sensing of nucleic acids.¹⁸⁷ The process that enables this effective sensing is the “surface plasmon resonance effects in metal nanoparticles” and this directly enables label-free detection of biomolecular interactions. Next to Au and Ag NPs, quantum dots are the nanomaterials used adequately for biochemical sensing. The notable property of these materials is their ability to sense from multiplexed samples (that is they can detect multiple analytes in biofluids simultaneously). These materials work based on tunable fluorescence and have exceptional photostability.^{188,189} These examples highlight the capabilities of nanomaterials in providing high sensitivity and selectivity to sensors. When such nanomaterial based biochemical sensors are integrated with AI-techniques, they enable automated optimization of sensor parameters and real-time compensation for environmental effects that could affect measurement accuracy.^{190,191}

5.2 Flexible electronics and biocompatible materials

A primary requirement for non-invasive sensing is the necessity of a conformal contact with the sensing platform. Traditional electronics based sensing systems are rigid, and thus, are poorly suited for wearable applications. The reason for their poor suitability relies in the fact that such rigid devices offer mechanical mismatch with the biological tissues and thus, user comfort becomes a concern. Biocompatibility and bio-comfort for a biosensor system are enabled by developments in flexible and stretchable electronics powered by nanotechnology. These flexible and stretchable materials play a significant role in creating comfortable, skin-conforming biosensors that can function as desired without disturbing normal working conditions.^{192,193} There are numerous materials that offer realization of flexible electronics which include organic semiconductors, conducting polymers, and hybrid materials. These materials when used for realization of sensors enable sensing systems that can bend, stretch, and conform to the curved surfaces of the skin without losing their functionality.^{194,195} In addition, these materials are biocompatible which enables their prolonged use in direct contact with the skin as well as biofluids without causing major effects.^{196,197} Next to organic semiconductors, polymeric materials are widely investigated for realization of flexible electronics. Materials like PDMS, polyurethane, and hydrogels have good mechanical flexibility as well as provide excellent biocompatibility which is required for wearable applications.^{198,199} After organic and polymeric materials, the use of self-healing materials has been widely studied. These materials are capable of autonomously repairing minor damages caused by stretching, bending or environmental factors. Integration of such self-healing materials to wearable sensors potentially extends the device life-time and at the same time maintains their functionality despite mechanical

stress.^{200,201} Furthermore, biodegradable materials are also used for biosensor realization wherever temporary monitoring is needed and a device removal after monitoring is impractical. Such a situation arises in the case of monitoring implants or monitoring a surgical site.^{203–205} AI-assistance for monitoring sensor performance and properties could enable early detection of performance degradation and potential damages with the sensing system which results in triggering the self-healing process or alerting the users about device replacements.²⁰² Surface modifications and coatings can further improve biocompatibility and reduce the risk of inflammatory reactions or device rejection. Thus, these techniques form the topic of the next discussion.

5.3 Surface functionalization and bioconjugation

The required selectivity and sensitivity of a biosensor could be achieved with surface functionalization techniques. Surface functionalization involves modifying the sensor surface such that the sensor can recognize and bind to the target analyte alone. This helps in minimizing non-specific interactions from the bio-environment which when left unnoticed may cause false outputs.²⁰⁶ Such a surface functionalization is achieved by various bio-conjugation strategies which help immobilize recognition elements like antibodies, aptamers, enzymes, or synthetic receptors onto sensor surfaces.^{207,208} Self-assembled monolayers that result from spontaneous self-assembly of molecules provide a promising solution for surface modification. Such a SAM exhibits precise control over surface chemistry and molecular orientation. SAM-based surface modification offers a promising platform for biochemical sensing.^{209–211} Furthermore, surface modifications by hydrogel coatings and polymer brushes are capable of creating biocompatible interfaces that resist non-specific analytes in the sensing environment while showing affinity towards the target analyte.^{212,213} In addition to coatings, the use of stimuli-responsive materials enables switchable or tunable sensors with advanced functions.^{214–216}

The introduction of AI-based learning techniques helps in optimizing surface modification strategies. AI algorithms are capable of analyzing correlations between surface chemistry, binding kinetics and sensor performance which could be used for obtaining the most optimal functionalization possible.^{217,218} In addition, ML algorithms can be used for predicting the optimal surface density of recognition elements which could facilitate a feasible surface modification that minimizes bio-fouling. Similarly, AI can assist in designing multiplexed sensing systems that are capable of sensing multiple analytes simultaneously without cross-interference. Furthermore, automated material characterization techniques powered by AI algorithms can provide real-time feedback on surface quality and stability. Such feedback enables predictive maintenance and quality control in sensor manufacturing.^{219,220}

5.4 Miniaturization and integration challenges

Device miniaturization poses numerous technical challenges and these challenges should be addressed appropriately in

order to use the biosensor system for practically wearable or implantable applications.^{221,222} Primary limitations with device miniaturization include (i) increase in signal-to-noise ratio with decrease in device size, (ii) power consumption and thermal management becomes more significant. The micro-electromechanical systems (MEMS) technology could provide reliable sensor fabrication. And thus, MEMS techniques are adequately used for the fabrication of sensors with sub-millimeter dimensions while maintaining high performance.^{223,224} MEMS techniques for miniaturizing biosensors include (i) surface micromachining – patterning, micromachining, etching, (ii) bulk micromachining – dry etching and wet etching, (iii) LIGA techniques – lithography, electrodeposition and molding.²²⁵ However, integration of multiple sensors, signal processing units, powering modules, wireless communication modules and power management systems within a single unit involves numerous challenges.^{226,227} System-on-chip (SoC) techniques can provide a reliable solution for this integration issue. Such systems can integrate multiple units in a single substrate, thus reducing size and power consumption. In addition, SoC can offer reliable functions. Furthermore, three-dimensional integration techniques enable vertical stacking of numerous functional layers, which helps in reducing the overall device footprint.^{228,229}

A major challenge with integrated systems is their power management. This challenge is more pronounced in the case of miniaturized biosensor systems that are used for continuous health monitoring applications.^{230,231} In order to address this challenge numerous energy harvesting techniques were investigated with the aim of making the biochemical sensing system a self-powered one that does not require battery replacement. Energy harvesting from heat, electromagnetic fields, and body motion has been adequately researched in this regard. Furthermore, the combination of ultra-low-power electronics with intelligent power management algorithms enables optimizing sensor operation with usage pattern and measurement requirement. This facilitates extended battery life and optimal power management.^{232,233} Specifically, AI methods play a crucial role in power optimization by (i) predicting measurement requirements, (ii) adjusting the sensitivity of the sensor based on signal quality and (iii) managing communication schedules. These monitoring methods and modifications minimize energy consumption while maintaining adequate data quality.^{234,235} Thus, the material requirements for efficient biochemical

sensing are addressed by the use of nanomaterials, and tuning and enhancements in each stage are addressed by AI integration. The computing platform used for data analytics will be covered in the next section.

In summary, the impact of nanomaterials and nano fabrication techniques in biochemical sensing systems could be recognized across four major topics. They are nanomaterials (NMs) used directly as recognition elements, functional coating, which directly determines the sensitivity of the biochemical sensor, flexible substrates for device realization enabled by NMs in 3D forms, films and composite forms, and device miniaturization from nano fabrication techniques that enable bio-comfort and sensing quantities which are not possible with conventional devices. A detailed discussion of the recent work addressing these topics is presented in the above section while a summary of the presentation is given in Table 3 below.

6 On-device processing for ensuring security

On-device data processing enables (i) storage of sensed data as well as (ii) immediate analysis of sensed data, (iii) faster computing, (iv) data security, and (v) autonomous operation. These requirements could be met with the help of edge computing. Edge-based architecture offers promising solutions for biochemical sensing applications. However, edge computing systems become inefficient when input data become immense and in such cases a hybrid edge-cloud system is introduced. In addition, in federated learning techniques the model alone is shared among different distributed locations and they are trained on data that are safe in the particular node. This technique offers the required analytics in spite of ensuring data security and data diversification. Recent research studies highlighting the benefits of these techniques in biomedical analytics are discussed in the below subsections.

6.1 Edge computing and real-time analytics

Biochemical sensing devices when integrated with edge computing capabilities offer numerous benefits which include (i) on-device processing, (ii) computations on sensed data to occur at the network's edge instead of on a central cloud or server, (iii) ability to work with minimal bandwidth requirements, (iv) minimal to zero cloud system access, (v) reduced

Table 3 Impact of nanomaterials and fabrication techniques on biochemical sensing

S. no.	Description	Materials	Outcomes
1	Nanomaterials in biochemical sensors	Carbon nanotubes, graphene, and carbon quantum dots, metallic nanoparticles, conductive polymers	Improved sensitivity and selectivity, device miniaturization, single molecule sensing, multiplexed sensing
2	Surface functionalization	Hydrogel coatings and polymer brushes	Modify target binding capabilities, antifouling, and antibacterial actions
3	Flexible electronics	PDMS, polyurethane, and hydrogels	Conformal contact with the skin, biocompatibility for long term use
4	Device miniaturization	Surface micromachining, LIGA techniques	Bio comfort, reduces power consumption, enables self-powering

latency, (vi) improved data security. These capabilities provide real-time monitoring of sensed data, with remote access capabilities and various data analytics that enable predictions and forecasts on medical data.^{236,237} Such edge computing architecture can be realized with the help of modern microprocessors and specialized AI chips. These devices can provide substantial computational capabilities within the limitations posed by wearable devices and thus could be reliably used with medical wearables.^{238,239} As data analytics with AI techniques facilitate numerous interpretations, ML algorithms based on low-power computing architecture and model optimization techniques can be directly deployed on biosensor devices with edge computing architecture.²⁴⁰ ML model optimization is carried out by model quantization, pruning, and knowledge distillation methods. These methods reduce the computational and memory requirements of AI models in spite of maintaining the required performance levels.²⁴¹ Furthermore, the use of specialized neural processing units (NPUs) and AI accelerators designed for edge applications can provide excellent computing platforms for ML workloads with reduced power consumption.^{242,243}

Edge computing enabled real-time processing of biosensor signals was demonstrated to be effective in identifying cardiac arrhythmias, hypoglycemic episodes, or exposure to toxic substances. For instance, ML pipelines deployed on edge computing based Open Neural Network Exchange (ONNX) formats were demonstrated to be effective in classifying Atrial Fibrillation (AFIB), Normal Sinus Rhythm (NSR), and Atrial Flutter (AFL).²⁴⁴ Similarly, edge computing techniques were used for the prediction of hypoglycemic events.²⁴⁵ Furthermore, AI algorithms on-device can provide patient-specific analysis. This analysis adapts to individual users with time and improves the accuracy of predictions and reduces false alarms. In addition, edge-based processing facilitates secured data processing (as all data are processed locally without transmission to external servers), which is a primary requirement while processing sensitive health data.^{246,247} In addition, real-time analytics with edge based computing were demonstrated to be time efficient which is critical for healthcare monitoring where delay could create adverse consequences.^{248,249}

6.2 Cloud-edge hybrid architecture

Though edge computing techniques offer numerous benefits when it comes to a complex analysis, where large-scale data storage and model training are required cloud-based models excel over edge computing techniques.^{250,251} Cloud computing can facilitate services like epidemiological monitoring, population-level analysis, as well as collaborative learning from multiple devices and users.^{258,259} Specifically, cloud-based systems could offer valuable services with large-scale data. Such large-scale data analytics culminate the data from a huge community or a number of devices. Such analysis can identify patterns and trends in data that are not predictable from individual device data. In addition, cloud based services can facilitate software updates, model improvements, as well as distribute new analytic capabilities to other deployed devices in the network.^{260,261} Thus, hybrid architecture that combines the

benefits of edge as well as cloud based processing is used for achieving optimized results. Such hybrid architecture enables optimization of the trade-off between latency, computational capabilities and resource constraints. Such hybrid architecture uses edge based local processing to handle immediate analysis, emergency responses and depends on cloud-based sophisticated analysis when long-term monitoring, trend analysis, complex decision making from multimodal data, and forecasting are required.^{252,253} These hybrid systems use intelligent filtering techniques at the edge to reduce the bandwidth requirements and computing cost associated with cloud based analytics. Such data filtering and aggregation enable transmitting only relevant information to the cloud instead of sending raw sensor data.^{254,255} Furthermore, ML models can be used to separate the data to be processed locally and those that need a cloud-based computation. This separation is achieved effectively with ML methods due to their ability to segregate data based on factors like emergency processing, quick response, complexity of the required analysis, and availability of computational resources. Specifically, the use of adaptive algorithms enables dynamic adaptation that adjusts the division of work between cloud and edge based on the present state of the system and the requirements.^{256,257} In such hybrid architecture, blockchain technologies were explored for ensuring data integrity and secure data sharing between edge and cloud platforms. Furthermore, these techniques facilitate maintaining user privacy and data ownership rights.^{262,263}

6.3 Federated learning for healthcare

Federated learning (FL) approaches enable training AI models with diverse data obtained from various institutions across the globe in a secure manner. FL methods enable the participating institutions to share only their model updates after training with their own dataset. Such a learning facilitates securing datasets locally and at the same time enables training models on diverse data.^{264,265} As the model updates are shared, a central server will aggregate these updates to improve the global model outcomes. Thus, models are collaboratively trained across multiple devices without centralization of raw data.^{266,267} Such FL models are highly beneficial in healthcare applications where data privacy is of paramount importance, and numerous modifications of FL frameworks have been proposed for healthcare applications. For instance, an FL approach combined with ring signature defense is demonstrated to prevent source inference attacks with FL based systems.²⁶⁸ Similarly, Privacy Protection Scheme for Federated Learning under Edge Computing (PPFLEC) was demonstrated to offer protection against equipment dropping, collusion attacks, as well as replay attacks.²⁶⁹ In addition to providing security, FL models offer development of more robust and generalizable models by enabling efficient model training by collaborative learning. This collaborative nature of FL models aids in addressing limited data problems which prevail in healthcare applications.^{270,271}

Future models based on FL methods will be based on advanced techniques that address challenges related to data heterogeneity and communication constraints, and at the same

time ensure device reliability across healthcare environments.^{272,273} Furthermore, FL models can be customized and used by individual researchers. In such cases, the FL models that are basically population-level learned models are customized for individual users (data) for achieving the required performance. Furthermore, secure aggregation algorithms (nw258) ensure that individual model updates are not used for tracing back sensitive information.^{274,275} The inclusion of federated learning architecture with biosensor systems facilitates time-to-time model adaptation while maintaining the highest standards of data privacy and security.^{276,277}

6.4 Emerging technologies for privacy-preserving analytics

As healthcare data are highly sensitive, they pose numerous risks associated with potential for misuse and unauthorized access.^{278–280} Conventional privacy protection techniques like data anonymization and access restrictions become insufficient under continuous monitoring applications as patterns in data could be potentially used to identify individuals or reveal sensitive information underlying them.^{281,282} Numerous security measures were proposed to address these challenges. One among them is the differential privacy technique which facilitates adding calibrated noise data or analysis results along with the original data. Such a method provides mathematically guaranteed privacy protection and meaningful analytics, and limits the ability to extract information about a specific individual.^{283,284} Next is the homomorphic encryption technique which facilitates computations to be performed on encrypted data as such without decrypting it. Such a technique ensures secure cloud computing by maintaining data confidentiality.^{285,286} Furthermore, secure multi-party computation protocols enable collaborative analysis across multiple parties without revealing individual datasets.^{287,288}

However, implementation of privacy-preserving analytics in biosensor systems demands a careful analysis and consideration of trade-offs between the analytical utility of a system and privacy protection.^{289,290} Thus, advanced hybrid techniques that combine federated learning, differential privacy and secure computations together could facilitate security across different layers of the system enabling layered privacy protection. Furthermore, ML methods can be adapted to minimize extraction of sensitive information as well as maintain clinical utility.^{291,292} Imparting the understanding of user access control and transparency mechanisms enables individuals to know how their patient data are being used. This enables them to impart control over their privacy preferences.^{293–295} Thus, development and deployment of privacy-preserving analytics become essential for building trust and enabling widespread adoption of AI-driven biosensor systems.^{296–298}

7 Clinical applications and validation studies

Intrinsic to their capabilities, AI enabled biochemical sensing systems are used across the length and breadth of healthcare services. The three major applications include (i) chronic

disease management – alerting users and caretakers by timely intervention, (ii) early disease detection – by predictive analytics and (iii) personalized medicine – the learning capabilities of AI tools enable adapting to user-specific patterns. These topics will be discussed with examples from recent research work in the below subsections.

7.1 Chronic disease management

AI powered non-invasive biochemical sensors are emerging as promising solutions for chronic disease management, by offering (i) continuous monitoring, (ii) early anomaly intervention and (iii) personalized treatment optimizations.^{299,300} One of the most successful applications is the continuous glucose monitoring (CGM) systems used for diabetes management. Modern CGMs are powered by ML algorithms and enable predicting glucose trends and detect impending hypoglycemic or hyperglycemic events. Furthermore, certain devices offer personalized medicine, that is, delivering insulin based on patient needs.^{301,302} Next to diabetic management, biosensors find vast application in cardiovascular disease (CVD) monitoring. Recently developed wearable sensors are capable of detecting cardiac arrhythmias earlier. Furthermore, when enabled with data analytics for making trend predictions, these devices can monitor blood pressure trends as well as assess cardiovascular risk factors.^{303,304} Specifically, AI powered CVD monitoring devices can analyze complex patterns inherent in physiological data and identify cardiovascular risk factors like atrial fibrillation, early signs of heart failure, and other cardiac functions even before clinical symptoms appear. Studies reveal that AI-enhanced CVD monitoring devices enable benefits like (i) reduced healthcare cost, (ii) improved patient benefits, (iii) early disease intervention, and (iv) prevention of acute episodes.^{305,306} The third most researched and biosensor powered disease management is related to respiratory disease. Wearable respiratory sensors could facilitate enhanced respiratory disease management through breath analysis.^{307,308} Such wearable sensors based on breath analysis can (i) monitor lung function parameters, (ii) detect exacerbations of chronic obstructive pulmonary disease (COPD) or asthma, as well as (iii) assess treatment effectiveness. Furthermore, when powered by AI-based analytics, breath analysis based wearable monitoring devices are capable of analyzing breathing patterns, coughing sounds, as well as identify volatile biomarkers to provide comprehensive respiratory health assessments.^{309,310} Studies demonstrate that AI driven respiratory monitoring wearables reduce hospital admissions and improve the quality of life of patients with chronic respiratory conditions.^{311,312}

7.2 Early disease detection

Early disease diagnosis – often characterized by the ability to detect diseases even before clinical symptoms appear – presents numerous benefits in healthcare applications ranging from individual wellness to preventing a disease outbreak.^{313,314} Considering individual wellness, diagnosing diseases in early stages enables (i) introducing treatment procedures when the disease is most treatable, (ii) probability of obtaining a cure is

more, (iii) reduced healthcare cost. For example, cancer if detected in early stages is curable in most cases. Non-invasive breath analysis has shown promising results in diagnosing lung, breast, and colorectal cancers from VOCs in exhaled breath.^{315,316} Such a diagnosing ability could facilitate early treatment interventions which may cure the disease altogether. For instance, AI-enabled biochemical sensing has demonstrated promising potential in diagnosing various neurological diseases in their early stages. For instance, eye tracking from video by using CNN algorithms, SVMs and linear mixed models have shown promising results in diagnosing Parkinson's disease, Alzheimer's disease and various other neurodegenerative conditions.³¹⁷ Similarly, various non-invasive procedures were demonstrated to be helpful in diagnosing various neurodegenerative conditions.³¹⁸ Specifically, speech pattern analysis, gait monitoring, and cognitive assessment by using wearable devices and non-invasive diagnosis can provide early indicators for numerous neurological decline. ML algorithms can identify and detect even very subtle changes in biomarker patterns that may precede clinical diagnosis.^{319,320}

A significant and crucial role of AI-assisted non-invasive diagnosis is in the diagnosis of infectious diseases. Such an advancement has gained attention particularly following the COVID-19 pandemic. Non-invasive measurements and diagnosis became a mandatory requirement during pandemic outbreaks.^{321,322} Specifically, breath analysis, saliva testing, non-contact fever measurements and respiration monitoring for symptoms were explored adequately for early detection of infectious diseases during the pandemic. In addition, AI algorithms that can analyze patterns in physiological data were demonstrated to identify pre-symptomatic infections early which support public health surveillance efforts.^{323,324} Clinical validation studies on AI-assisted non-invasive biochemical sensing have shown promising results for numerous infectious disease detection applications. However, challenges persist with respect to achieving sensitivity and specificity of the predictions for validating their use for widespread clinical screening applications.^{325,326}

7.3 Personalized medicine applications

Next to early disease diagnosis, AI-powered bio-sensors are used for precise and patient-specific drug delivery. Wearable biochemical sensing devices could continuously monitor the analyte of interest which is usually a disease biomarker, and based on analysis outcomes could deliver a precise amount of the drug. However, the drug-delivery could be automatic or semi-automatic.^{327–329} A fully automatic drug delivery system will be capable of monitoring the pharmacokinetics (drug levels and metabolites) in real-time with non-invasive sensors and optimize drug dosing. Such an approach is a promising solution for drugs with narrow therapeutic windows or significant inter-individual variability in metabolism. However, a semi-automatic drug delivery system demands a physician intervention before making changes to drug dosing.^{332,333} Numerous AI-assisted drug monitoring devices were investigated in the literature. For instance, non-invasive glucose sensing using

microneedle patch type systems and using glucose levels to optimize insulin delivery was demonstrated to offer a precise and closed loop control in insulin delivery systems. This system used a microcontroller to hold a rule-based algorithm to quantize glucose levels and insulin requirement and included smart phone based monitoring.³³⁰ Even monitoring of lithium drug for bipolar disorders is demonstrated in the literature.³³¹ AI-based machine learning algorithms can predict individual drug responses and identify optimal treatment plans based on real-time data from biosensors. Clinical studies validated improvements in treatment outcomes and reduction in adverse effects that result from personalized drug dosing based on continuous biomarker monitoring.^{334,335} In addition to monitoring disease biomarkers for precise drug delivery, non-invasive continuous monitoring could be used for monitoring various factors like stress hormones, sleep quality, physical activity and metabolic parameters to gain insights on health patterns and guide/provide assistance in personalized lifestyle interventions.^{336,337} AI algorithms play an important role in understanding correlations between lifestyle factors and health outcomes in order to provide recommendations on stress management, exercise as well as diet.^{338,339} Furthermore, integration with digital therapeutic platforms enable delivery of personalized healthcare interventions based on real-time data for potentially improving long-term health outcomes and reducing chronic disease burden.^{340,341}

7.4 Clinical trial integration

The use of AI-assisted biochemical sensing systems extends beyond the treatment and therapeutic interventions and revolutionize drug developments by facilitating easiness of clinical trials. With the integration of smart biochemical sensing systems, continuous, objective measurements of treatment effects and patient responses have become faster and easier, which reduces the lab-to-clinic time.^{342,343} Furthermore, smart biochemical sensing driven clinical trials enable continuous data on drug efficacy, safety, and pharmacokinetics throughout the trial period which may not miss important changes or variations in treatment responses. This is not the case with traditional clinical trials which rely on periodic assessments where there are chances of missing important variations.^{344,345} Such a continuous sensing is assisted by digital biomarkers that can provide more sensitive and objective measures of treatment responses compared to traditional assessments.^{346–348} AI algorithms capable of identifying even subtle changes in physiological patterns can create a digital twin of a patient from time-to-time and verify treatment effects and drug effects. Such an approach has the capability to reduce trial duration, improve statistical analysis and enable earlier detection of safety signals.^{349,350}

Remote monitoring capabilities of smart non-invasive biosensor systems reduce patient burden as they nullify the need for frequent clinical visits.^{353,354} At the same time, the decentralized nature of trial sites opens up possibilities of reaching a broader patient population. This certainly removes the geographical barrier for trial participation. Furthermore,

ML algorithms ensure data quality and aid in identifying potential issues in real-time enabling rapid medical intervention when needed.^{355,356} The primary driving force for such remote monitoring technologies in every dimension of healthcare is the COVID-19 pandemic. The pandemic has accelerated non-invasive disease diagnosis, remote monitoring for healthcare delivery as well as clinical trials. It demonstrated the feasibility as well as the potential benefits of such technologies, in particular, for future drug development efforts.^{357,358} In spite of these benefits, there lie the challenges for its ethical use and thus regulatory agencies are recognizing the value of digital biomarkers and started developing guidelines for their clinical trial applications.^{351,352}

8 Challenges and limitations

Due to the capabilities of biosensors, there exist numerous challenges that hurdle their effective design and deployment. These challenges exist along three verticals namely (i) technical – biocompatibility of devices for medical use, toxicity assessments, (ii) regulatory – are they tested safe? what needs to be tested to ensure safety? and (iii) user specific – concerns regarding security and responsibility of user data. The components of a biosensor-based healthcare device, along with the methods used to address various challenges that exist in realizing each of its components, are presented in Fig. 7. These design changes at the technical level enable overcoming the limitations in other levels too. Fig. 7 represents the various measures taken at the technical level, to address the challenges that are posed on a non-invasive biochemical sensing device.

8.1 Technical challenges

In spite of the capabilities of non-invasive sensing, there exist numerous technical challenges for the widespread adoption of smart non-invasive sensors for clinical use.^{359,360} For instance, issues related to sensor calibration and drift compensation hurdle the long-term use of biosensors. This issue results from changes in the sensitivity of the sensing element with time which might result from numerous factors like (i) biofouling, (ii) biodegradation of the recognition element, (iii) environmental factors.^{361,362} Though AI-driven calibration adjustments

could compensate for drifts and facilitate the required accuracy, maintaining accuracy over an extended time period remains a challenging task. Apart from sensor degradation, issues that affect biochemical sensing are cross-sensitivity and interference from non-target analytes. The biofluid considered for sensing the analyte contains numerous other compounds, which may interfere with the sensor measurements. Furthermore, the composition of the biofluid varies between individuals as well as time. Such interferences hurdle selective detection of analytes in complex biological matrices.^{363–365} Though ML algorithms can compensate the sensitivity loss, advances in sensor selectivity and signal processing techniques could drive a robust sensing platform.^{366,367}

Next to the sensitivity of the biosensor system the factor that hurdles the use of non-invasive sensing systems is the power demands of the device. Continuous monitoring applications are operated over an extended time period and thus (i) inclusion of complex computations on-device and (ii) frequency of measurements taken are both limited by the power constraints of the device. Thus, power consumption and battery life pose limitations on wearable as well as implantable biochemical sensing systems.^{368,369} In order to tackle this limitation, various energy harvesting techniques were investigated while designing wearable sensor systems. However, the energy harvested from such smart materials is very meagre and not sufficient for smart biochemical sensing applications.^{370,371} In order to address these limitations concerning power management, design of ultra-low-power electronics along with intelligent power management strategies is essential for the practical realization of smart biochemical sensing and continuous monitoring systems. Manufacturing variability and quality control measures present challenges for producing reliable and reproducible biosensors at scale.^{372,373}

8.2 Regulatory and ethical considerations

Conventional medical device regulations were all framed considering rule-based computing methods but AI powered medical sensing devices are capable of learning continuously from the input given to them and adapt to the changes over time. Thus, the existing medical device regulations become highly insufficient to validate the devices based on AI

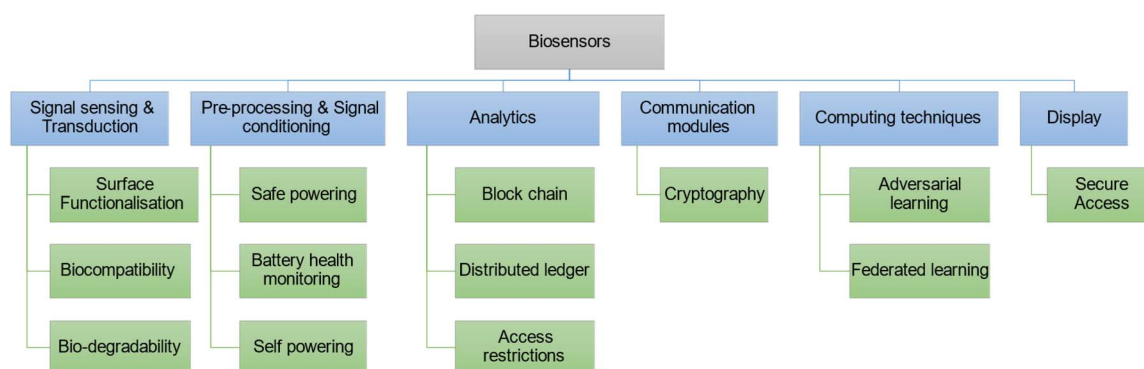


Fig. 7 Techniques adapted to address challenges across a biosensor system.

techniques. Specifically, the ethical questions regarding “the responsibility for AI driven risk” that arise with the use of AI-driven solutions for highly sensitive medical applications present unique challenges while framing regulations for device validation.^{374,375} This necessitated regulatory agencies to develop new frameworks for approving AI-enabled medical devices for clinical use. However, the approval pathways are even more complex and are time-consuming which resulted in the increase of lab-to-clinic time.^{376,377} In addition, clinical validations become mandatory for AI acceptance, however, they poses challenge inherent to their demand to demonstrate both analytical and clinical validity across diverse populations.^{378,379} Furthermore, regulatory agencies demand evidence that supports algorithm robustness, bias mitigation as well as performance monitoring capabilities which becomes highly difficult with AI predictions.^{380,381}

Ethical considerations surrounding AI-powered wearable healthcare technologies include uncertainty with respect to numerous factors like (i) algorithmic bias, (ii) fairness among predictions, (iii) equity in healthcare delivery.^{384,385} These concerns are fueled by the “Black Box” nature of the ML algorithms. This raises questions regarding model explainability and clinical interpretability. Furthermore, considering training data, usually they represent a particular population and may not include the whole global community. This leads to difference in the performance of the model across different demographic groups. However, ensuring equitable access to smart medical technologies and preventing creation of digital health divides are important ethical challenges.^{386,387} The next most inevitable demand is the questions regarding the procedures to obtain informed consent from patients before using AI-driven systems for their healthcare solution. The primary questions that arise include (i) what a patient needs to understand about the complex algorithms, (ii) how consent should be obtained for systems that may evolve over time?^{388,389} Furthermore, the decentralized nature of the present regulatory considerations demands navigation of different regulatory requirements across jurisdictions in order to prevent the digital divide. These requirements add to the complexity of product development and commercialization efforts.^{382,383}

8.3 Data privacy and security

Traditional medical tests and PoC diagnostic devices generate discrete data that reveal the health condition of an individual at the time of testing, whereas AI-driven smart devices create a detailed record of the physiological and behavioral patterns of individuals. This nature of the AI-driven biochemical sensing devices creates significant privacy and security challenges, where these data can be used to infer sensitive information about individuals.^{390,391} Such data breaches on continuous health monitoring data could have severe consequences on individuals which include discrimination in employment, insurance, or social contexts.^{392,393} Some examples of such cybersecurity threats connected to biosensor systems include (i) unauthorized access to sensing system – which is highly dangerous resulting in chances of altering device characteristics, (ii) data interception during

transmission – which enables identifying individual user identity, (iii) attacks on cloud based analytics – which reveals trend in medical conditions of an individual.^{394,395} These treats depict the potential possibilities for malicious actors to manipulate sensor data. Furthermore, AI algorithms could be used to trace back patient data which pose risk to patient safety and system reliability. These risks demand implementation of robust security measures without compromising device usability or performance and this presents numerous challenges.^{396,397} Privacy-preserving techniques such as differential privacy and federated learning offer promising solutions, however they demand an optimal tradeoff between privacy protection and clinical utility.^{398,399}

Next to data privacy is the issue of data ownership and responsibility, in a single term “accountability”. This issue represents a complex issue in the context of AI-driven biosensor systems.^{400,401} Questions like (i) who owns the data? (ii) how can the data be used? (iii) what rights does an individual have to control or delete their data? remain unanswered in the context of AI-assisted healthcare monitoring. Individuals' rights on their own data and the value of aggregated health data for research and public health should be balanced (considering the ethical context).^{402,403} These issues are fueled by the non-availability of global and standard regulations and privacy laws across jurisdictions. Developing transparent data governance frameworks that respect individual rights and at the same time enable beneficial use of health data remain an ongoing challenge.^{404,405}

8.4 Standardization and interoperability

Global standards related to biosensor technologies, data formats and AI algorithms present numerous barriers to the widespread adoption and clinical integration of AI-powered biochemical sensing systems.^{406,407} Some of the primary issues include the following.

(1) Data formats, communication protocols and analytical approaches vary between manufacturers which result in (i) limiting interoperability between devices and systems, (ii) healthcare providers resist the use of multiple incompatible systems, (ii) patients will not be able to make a choice between monitoring solutions.^{408,409}

(2) It is difficult to integrate electronic health records and clinical decision support systems with biosensor data.^{410,411}

(3) Comparing the quality and performance of various AI algorithms and medical devices becomes difficult due to the unavailability of standards (on both performance metrics and testing data). Furthermore, it is difficult to identify the best practices adopted for satisfying the requirements.^{414,415}

(4) Though industrial organizations and regulatory agencies are working to develop standards for AI-enabled medical devices and digital biomarkers, the progress has been slow due to the rapidly evolving nature of the technology.^{412,413}

The following measures are recommended for consideration while developing standard regulations:

(1) Development of standard (i) evaluation protocols, (ii) performance metrics and (iii) reference datasets for analyzing

AI-driven biosensor applications. This could support in accelerating the approval processes.^{416,417}

(2) Standards should be framed considering international collaboration and industry cooperation in order to ensure development and implementation of effective standards.^{418,419}

9 Emerging technologies and future directions

9.1 Quantum sensing for biomedical applications

Quantum sensing represents a promising technology that provides dramatically higher sensitivity and precision to non-invasive biochemical sensing systems.^{420,421} Quantum sensors are based on quantum mechanical phenomena which include superposition, entanglement and quantum interferences to achieve measurement sensitivities that exceed classical limits. Quantum sensing technologies could detect single molecules, measure weak magnetic fields from neural activity as well as provide unprecedented spatial resolution for biological imaging.^{422,423} Nitrogen vacancy (NV) centers in diamond are emerging as promising quantum sensors for biomedical applications.⁴²⁴ The quantum defects in the diamond atom can be used for detecting magnetic fields with extraordinary sensitivity. This facilitates applications such as single-cell magnetometry, detection of magnetic nanoparticles, and analysis of biomolecular interactions.^{425,426} Next to NV vacancy in diamond, optically pumped magnetometers can represent another class of quantum sensors reliably used in biomedical applications. These sensors offer high-sensitivity magnetic field detection (without superconducting electronics) which is used in brain imaging and cardiac monitoring applications.^{427,428}

AI integration with quantum sensors enables (i) optimizing sensor operation in real-time, compensating for noise and drift from the environment, (iii) extraction of meaningful biological information from complex sensor data. These optimizations enable efficient biomedical measurement and analysis.^{429,430} Such ML approaches could accelerate the development of new quantum sensing protocols as well as facilitate automated discovery of quantum sensing applications.^{431,432} Though quantum sensing technologies are still in their infancy, they offer tremendous opportunities for next-generation biochemical sensing capabilities and hold promise for transforming medical diagnostics and biological research.^{433–435}

9.2 Next-generation AI algorithms

Similar to material advancements, AI advances for the future offer promising capabilities with biosensor data analysis and interpretation.^{436,437} Some of the promising techniques include the following.

(1) Transformer architecture: these models are originally developed for interpreting text thus enabling natural language processing. Such models when adapted for interpreting patterns from timeseries biomedical data were demonstrated to provide excellent results. These models use attention mechanisms and were successful in retrieving the most relevant features from long range dependencies in sensor data.^{438,439}

(2) Graph neural networks: GNN based models could extract patterns from multi-modal data effectively and thus offer promising solutions for obtaining patterns from complex biological systems. These networks can model relationships between various biomarkers, sensor data, and physiological processes, and provide comprehensive and interpretable analysis.^{440,441}

(3) Neuro-symbolic AI: these methods combine neural networks with symbolic reasoning. Such methods are used for imparting explainability to AI models and thus enable the development of trustworthy biosensor analytics.^{442,443}

(4) Foundation models trained on large-scale biomedical datasets: such models offer the benefit of knowledge learned from diverse biomedical datasets and they can be tailored for a specific analyte sensing. These models offer powerful solutions for biochemical sensing applications.^{444,445}

(5) Few-shot learning and transfer learning: these techniques could offer promising solutions to biomedical applications which normally are affected by limited training data problem. By rectifying the limited data problem these models enable rapid deployment of AI-driven biochemical sensing applications.^{446,447}

(6) Continual learning algorithms: these models are capable of learning from the inputs used for predictions and adapt model weights to the changing inputs and thus, offer improved efficiency with time.^{448–450}

9.3 Integration with digital health ecosystems

The future of AI-driven non-invasive sensing systems lies in their integration with a comprehensive digital health ecosystem where devices, data, as well as healthcare services co-exist. Interoperable platforms are capable of aggregating data from multiple sensors, electronic health records, genomic information, and lifestyle factors that facilitate gaining more insights into health and disease.^{451,452} Such a digital health ecosystem will facilitate

(1) Treatment optimization and personalized predictions by updating the digital twin of individual patients with continuous real-time biosensor data.^{453,454} As AI-driven biochemical sensing systems can provide remote monitoring and accessing capabilities, they can support (i) remote consultations and enable continuous monitoring between clinical visits, (ii) extend expert healthcare support even to the underserved population^{455,456}

(2) Integrating NLP capabilities with biochemical sensing systems offers smart health care assistance, personalized health guidance as well as medication reminders.^{457,458}

(3) A secure, transparent and patient controlled health data ecosystem could be achieved with the introduction of blockchain and distributed ledger technologies. This enables individual privacy as well as data ownership rights.^{459,460}

(4) Compensation for sharing health data for research efforts could be made effectively by smart contracts.^{461,462}

In spite of these benefits, integration of biosensors with digital health ecosystems demands attention to standards, interoperability, and data governance in order to ensure that an

efficient potential is harvested from these technologies while maintaining data privacy and individual rights.^{463,464}

9.4 Sustainability and environmental considerations

The next dimension of advancements deals with the development of biosensors that offer sustainability and have very minimal to zero environmental impact.^{465,466} Sustainable designs that incorporate reparability, recyclability, and environment-friendly materials while developing biosensors could reduce concerns about electronic waste, resource consumption and disposal processes.^{467,468} Materials and methods adapted for achieving this goal include the following.

(1) Use of biodegradable materials that can easily dissolve in biological environments offers promising, environment-friendly solutions for temporary monitoring applications.^{469,470}

(2) Energy harvesting techniques reduce the necessity to depend on batteries and enable reliable operations.^{471,472}

(3) Green manufacturing interventions that facilitate reduced energy consumption and minimize waste are recommended for sustainable production of biosensor technologies.^{473,474}

(4) Life cycle assessments that enable assessing the environmental impact of biosensors and guide sustainable alternatives are frequently used.^{475,476}

(5) Circular economy principles are used for extending the useful life of biosensors.^{477,478}

In spite of these steps towards the development of a sustainable economy, the development of biosensors that provide promising health benefits along with minimizing environmental impacts remains challenging.^{479,480}

10 Market analysis and commercialization

10.1 Current market landscape

The biosensor market has experienced an unprecedented growth. The primary driving force behind this growth is the demand for development of (i) PoC diagnostic tools, (ii) continuous health monitoring devices, (iii) personalized drug delivery devices.^{481,482} As these devices could provide non-invasive, convenient and comfortable monitoring solutions facilitated by advances in wearable technologies, consumers show interest in using such devices.^{484,485} Due to their significance, the market value of smart wearable devices is valued at approximately \$70 billion USD by 2025.⁴⁸³ Development of smart wearables in the form of smart watches, fitness trackers, and specialized medical devices is further fueled by investments from technology companies.^{486,487} Specifically, Apple, Google, Samsung, and Fitbit are integrating biosensing capabilities to their consumer devices, which is driving market adoption and consumer awareness. Furthermore, these gadget giants are motivated to integrate biochemical sensors in their wearables driven by the success of continuous glucose monitoring systems from Dexcom and FreeStyle, as these systems have demonstrated the feasibility of realizing non-invasive health monitoring technologies.^{488–490}

10.2 Investment trends and funding

Apart from the normal technological development and investment trends, there is a huge change, convergence or unusual collaborative work force that is driving developments in the non-invasive sensing technology market globally. A simple investigation of these factors is presented below.

(1) In addition to investments from existing market players, investments from venture capital, government funding, and corporate investments are acting as the driving force for the development of smart biochemical sensing devices.⁴⁹¹ The global digital health funding was \$17.8 billion in 2019, reached \$28.4 billion in 2020, and peaked at \$52.7 billion in 2021,⁴⁹² where a remarkable percentage was allotted for the development of biosensor and monitoring technologies.

(2) Government initiatives like (i) the U.S. National Institutes of Health's Small Business Innovation Research (SBIR) program and (ii) the European Union's Horizon Europe program are encouraging incubation facilities and startups in digital health and biosensor research and development.^{493,494}

(3) The next factor is corporate partnerships and acquisition of small industries with biosensor capabilities by healthcare and technology companies.^{495,496}

Investments by major pharmaceutical companies in digital biomarker technologies to enhance their drug development and clinical trials, and technology companies have started acquiring healthcare capabilities. This convergence of healthcare industries, funding sources and AI technologies is the driving force for the present transformative paradigm.^{497,498} This change will be discussed in detail in the next subsection.

10.3 Key industry players

As examples for this prevailing convergence, (i) traditional medical device manufacturing companies such as Medtronic, Abbott, and Roche are developing advanced biosensor products and still continue to invest in next-generation technologies,^{499,500} (ii) technology companies like Apple, Google, Microsoft, and Amazon have started focusing on healthcare industries and development of AI driven healthcare solutions,^{501,502} (iii) emerging companies like Biolinq, Sano Intelligence, and Rockley Photonics are developing their own non-invasive monitoring technologies,^{503,504} (iv) academic and research institutions continue to contribute to biochemical sensing technologies and intellectual properties that drive industry innovations by rapid technological advancements, patent battles and strategic partnerships.^{505,506}

10.4 Commercialization challenges

In spite of this promising market growth, biosensor companies face challenges in transferring technology from the laboratory to the clinic.^{507,508} The primary challenges are the regulatory approval processes. With regard to AI-assisted medical devices, approval processes are complex, time-consuming, require extensive clinical validation and demand post-market surveillance.^{509,510} However, manufacturing scalability and quality controls present challenges for startups that transfer their

technology from research prototypes to commercial products.^{511,512} In particular, (i) manufacturing devices with consistent performance in a cost-effective manner demands significant investment in manufacturing and quality control, (ii) reimbursements and pricing strategies for novel biosensor technologies are either unclear or remain inaccessible in most of the cases,^{513,514} (iii) explainability constraints add to the complexity of approval terms.^{515,516}

11 Conclusions and research recommendations

11.1 Summary of key findings

The present review has examined the current state and future prospects of non-invasive biochemical sensing with AI-driven analytics, which is evolving rapidly to transform healthcare delivery. The pillars behind this emerging field are investigated and their contribution to non-invasive biochemical sensing devices are registered, that is, (i) advanced material science which paved the way for nanosized, highly sensitive, flexible, biocompatible, biodegradable, stimuli-responsive materials, (ii) nano technology which enabled device miniaturization and aided in achieving the unachievable, (iii) artificial intelligence which facilitated machines to learn and identify patterns from data, (iv) wireless communication which aided in remote monitoring capabilities. Furthermore, the clinical applications of smart sensing healthcare devices are investigated which revealed that non-invasive sensing helps in managing chronic diseases, early disease diagnosis and personalized medicine. Successful implementations of these applications include continuous glucose monitoring for diabetes management, cardiac monitoring for cardio vascular disease management and breath analysis for respiratory disease and cancer screening. Significant contribution to learning efficacy and security is made by federated learning and edge computing techniques.

11.2 Research gaps and opportunities

In spite of the advancements in non-invasive biochemical sensing, there exist numerous challenges that demand research intervention. This includes (i) the need for traditional testing methods to better understand the correlation between biomarker levels and disease states in various biofluids, (ii) the need for highly sensitive sensors and transducers that can detect biomarkers from low concentration complex biological matrices, (iii) algorithmic advances for realization of robust and clinically acceptable diagnostic systems – in terms of few-shot learning, domain adaptation, and explainable AI, (iv) the need for privacy preserving analytics which includes federated learning, and secure computation methods, (v) standards for data formats, performance metrics, and evaluation protocols that can facilitate interoperability and ethical regulatory approval.

11.3 Future research directions

Future research in the field of AI powered biochemical sensing should be prioritizing developments that focus on providing

reliable and robust non-invasive disease diagnosis, personalized predictions, disease forecasting, and improving security concerns. These objectives can be addressed by adopting the following measures.

- (i) Development of multi-modal sensing platforms that monitor multiple analyte simultaneously,
 - (ii) Integration of quantum sensing technologies that provide unprecedented sensitivity and spatial resolution,
 - (iii) Using advanced AI algorithms like transformer models, graph neural networks, and neuro-symbolic AI approaches that provide advanced biosensor data analysis and interpretation.
- The reliable use of AI-based biochemical sensing systems is highly dependent on ethical and regulatory considerations. Thus, ethical concerns should be addressed by
- (i) Collaborative research which includes engineers, computer scientists, clinicians, and regulatory experts,
 - (ii) Investment on enhancing clinical validation studies, regulatory science, and health economics research,
 - (iii) Paying attention towards sustainability, equity, and global health applications.

In summary, evolutionary developments in the field of AI-enabled biochemical sensing are promising that they have the potential to revolutionize the healthcare industry by enabling a new paradigm of preventive and personalized medicine.

Conflicts of interest

The authors declare that they have no conflicts of interest to report regarding the present study.

Data availability

No primary research results, software or code have been included and no new data were generated or analysed as part of this review.

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